

The unit of replication

a judge panel's n_{eff} ceiling is a statement about items, not judges

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Status. Theorems 1–7 are proved. Two claims I previously sent you are *withdrawn* and replaced: see §5. One side condition on the re-attribution is *unresolved on your data* and you are the only one who can close it: see §6.

1 Verdict, before the proofs

You asked a yes/no question — is “you’re paying for nine and getting two” defensible as an aggregate-reliability claim — and you are blocked on an answer, so here it is first.

1. **The number is right and you should not weaken it.** $\bar{\varphi}$ is *exactly* the between-item share of error variance, by an identity that needs no distributional assumption at all (Theorem 3). Kish’s n_{eff} is correct as written. It is strictly increasing in J and bounded by $1/\bar{\varphi}$.
2. **But it is a ceiling, and your 1.99 *is* the ceiling.** At $\bar{\varphi} = 0.49$ the ceiling is 2.04 and 1.99 is 97.5% of it. The honest headline is not “nine judges buy you two” — it is “**no panel size buys more than about two.**” That is a stronger sentence and it is free.
3. **The attribution does not survive.** τ^2 is the variance of a *per-item bias*: conditional on the item, every judge is wrong by the same amount, so the errors are not duplicated opinions, they are a shared offset. Replication over judges cannot remove an offset; replication over *items* averages it away. Hence Theorem 7: the only thing J can ever buy is $\sigma^2/(IJ)$, and the item count has no floor. **Kish’s design effect cannot tell redundancy from bias, and on alphaNLI it is reporting bias.**
4. **The exchange rate, done unconditionally.** At a fixed number of total judgments $B = IJ$, $\text{Var}(M) = (J(v + \tau^2) + \sigma^2)/B$ is *strictly increasing* in J , so $J = 1$ is optimal — no inequality to check, no dataset dependence (Theorem 10). This replaces the conditional version I sent you.
5. **The one thing that could still break it.** A judge main effect w_j (a judge harsh on everything) is *invisible to $\bar{\varphi}$* — annihilated by the per-column centring inside `np.corrcoef` — but it puts an ω^2/J term into $\text{Var}(M)$ that no number of items reduces (Theorem 14). **Your measured $\bar{\varphi} = 0.49$ cannot rule this out.** Measure ω^2 before the re-attribution goes in the headline; it is a one-way ANOVA on your judges’ column means and you already have the data. If ω^2 is non-negligible the honest word is “shared,” not “relocated.”
6. **Do not cut the panel.** The headline invites “nine $\rightarrow$ two,” saving 7/9 of spend. On the parameters implied by your own numbers that purchases a 35% *increase* in error variance, while the full $J = 9 \rightarrow \infty$ saving was only 10% of error variance to begin with (§8). The re-scoping is “spend it on items,” not “spend less.”

On your ESDOF question: errors, yes — but not “for consistency.” For identifiability (§7). And a warning: ESDOF and Kish are not two routes to one number. Their ceilings are $1/\bar{\varphi}^2$ and $1/\bar{\varphi}$, which on your panel is 4.16 versus 2.04 (Theorem 15).

2 The model, named

I am claiming a decomposition is *the* decomposition, in a field that is yours and not mine. So the model gets a name and an explicit falsifier.

Items are indexed $i = 1, \dots, I$ and judges $j = 1, \dots, J$. Write θ_i for the item’s true quality, X_{ij} for judge j ’s score of item i , and

$$E_{ij} := X_{ij} - \theta_i$$

for the *error*. For a binary verdict $E_{ij} = \mathbf{1}\{\text{judge } j \text{ is wrong on item } i\}$, which is the form your recipe consumes; for graded scores it is the residual, as your parenthetical already says.

Assumption 1 (Exchangeable conditional independence, **(ECI)**). There are i.i.d. item states $F_1, \dots, F_I$ (“everything about the item”: its difficulty, the ambiguity of its rubric, the misleadingness of its prompt) such that

- (i) conditionally on F_i , the errors $E_{i1}, \dots, E_{iJ}$ are independent;
- (ii) conditionally on F_i , the errors are identically distributed across j , so that $m(F_i) := \mathbb{E}[E_{ij} | F_i]$ and $s^2(F_i) := \text{Var}(E_{ij} | F_i)$ do not depend on j ;
- (iii) $\mathbb{E}[E_{ij}^2] < \infty$.

Write $\tau^2 := \text{Var}(m(F_i))$ and $\sigma^2 := \mathbb{E}[s^2(F_i)]$, and let $v := \text{Var}(\theta_i)$.

Part (i) is *yours and measured*, not assumed: you report (UID 742) that demeaning the errors per item collapses the cross-judge residual correlation to ≈ 0 . That is precisely (i). Part (ii) is exchangeability, and it is the part that can fail — §6 is devoted to exactly how.

Remark 1 (What (ECI) covers). (ECI) is deliberately weaker than a Gaussian random-effects model. Both of the following are special cases, and I verify the theorems in both:

- **(M1) additive Gaussian:** $X_{ij} = \theta_i + u_i + e_{ij}$ with $\text{Var}(u) = \tau^2$, $\text{Var}(e) = \sigma^2$. Here $m(F_i) = u_i$ and $s^2(F_i) = \sigma^2$.
- **(M3) conditional Bernoulli:** $E_{ij} | p_i \sim \text{Bern}(p_i)$ independently over j , with p_i random. Here $m(F_i) = p_i$, $\tau^2 = \text{Var}(p_i)$ and $\sigma^2 = \mathbb{E}[p_i(1 - p_i)]$.

The second is the honest model for your binary alphaNLI errors, and the theorems below hold in it verbatim — no normality is used anywhere.

Remark 2 (The falsifier, stated in advance). (ECI)(ii) fails if some judges are systematically harsher than others. The minimal extension is **(M2)**: $X_{ij} = \theta_i + u_i + w_j + e_{ij}$ with $\text{Var}(w) = \omega^2$. This is the most likely way everything below is wrong, and Theorem 14 says exactly what breaks and what does not.

3 Half one: the number survives

Theorem 3 ($\bar{\varphi}$ is the between-item error-variance share). *Under (ECI), for any $j \neq j'$,*

$$\text{Cov}(E_{ij}, E_{ij'}) = \tau^2, \quad \text{Var}(E_{ij}) = \tau^2 + \sigma^2, \quad \bar{\varphi} := \text{Corr}(E_{ij}, E_{ij'}) = \frac{\tau^2}{\tau^2 + \sigma^2}.$$

Proof. By the law of total covariance, for $j \neq j'$,

$$\text{Cov}(E_{ij}, E_{ij'}) = \underbrace{\mathbb{E}[\text{Cov}(E_{ij}, E_{ij'} | F_i)]}_{=0 \text{ by (ECI)(i)}} + \text{Cov}(\mathbb{E}[E_{ij} | F_i], \mathbb{E}[E_{ij'} | F_i]).$$

By (ECI)(ii) both inner conditional expectations equal the *same* random variable $m(F_i)$, so the second term is $\text{Cov}(m(F_i), m(F_i)) = \text{Var}(m(F_i)) = \tau^2$. By the law of total variance,

$$\text{Var}(E_{ij}) = \mathbb{E}[\text{Var}(E_{ij} | F_i)] + \text{Var}(\mathbb{E}[E_{ij} | F_i]) = \mathbb{E}[s^2(F_i)] + \text{Var}(m(F_i)) = \sigma^2 + \tau^2,$$

which is the same for every j by (ECI)(ii). Dividing gives the correlation. Finiteness of all three quantities is (ECI)(iii). $\square$

Theorem 3 is an *identity*, not a model fit: it is two applications of the total-(co)variance law and nothing else. Three consequences deserve naming.

Corollary 4 (The ceiling). *Let $n_{\text{eff}}(J) := J/(1 + (J - 1)\bar{\varphi})$. If $0 < \bar{\varphi} < 1$ then n_{eff} is strictly increasing in J , and for every finite J*

$$n_{\text{eff}}(J) < \lim_{J \rightarrow \infty} n_{\text{eff}}(J) = \frac{1}{\bar{\varphi}} = 1 + \frac{\sigma^2}{\tau^2}.$$

Proof. Write $p = \bar{\varphi}$. Treating J as a real variable, $\frac{d}{dJ} \frac{J}{1+(J-1)p} = \frac{(1+(J-1)p) - Jp}{(1+(J-1)p)^2} = \frac{1-p}{(1+(J-1)p)^2} > 0$. For the bound, $n_{\text{eff}}(J) < 1/p \iff Jp < 1 + (J-1)p = 1 + Jp - p \iff 0 < 1 - p$, true. The limit is $J/(Jp + 1 - p) \rightarrow 1/p$, and $1/p = (\tau^2 + \sigma^2)/\tau^2$ by Theorem 3. $\square$

Corollary 5 (What the ceiling is about). *The numerator of $\bar{\varphi}$ is $\tau^2 = \text{Var}(m(F_i))$, the variance of a function of the item state alone. The judge index j does not occur in it.*

That one line is the whole re-attribution, and it is why I think the flip is not a soft judgement call: the quantity that sets the ceiling is *definitionally* an item functional. There is no reading of $\bar{\varphi}$ on which the ceiling is a fact about judges.

Corollary 6 (τ^2 is a bias variance). *Conditionally on the item, the panel's mean error is $\mathbb{E}[\bar{E}_i | F_i] = m(F_i)$, independent of J . So $m(F_i)$ is a bias of the measurement on item i : it is not reduced by any amount of judging, and $\mathbb{E}[(\bar{X}_i - \theta_i)^2 | F_i] = m(F_i)^2 + s^2(F_i)/J$.*

This is the mechanism behind your own sentence “everyone finds the hard items hard.” The co-failure is not nine opinions collapsing into two. It is nine independent measurements of a quantity that is *offset*. Averaging independent measurements kills variance and leaves bias untouched — which is exactly what a ceiling looks like from the inside.

4 Half two: the attribution does not

Theorem 7 (The aggregate variance). *Let $M = \frac{1}{I} \sum_{i=1}^I \bar{X}_i$ with $\bar{X}_i = \frac{1}{J} \sum_j X_{ij}$, under (M1) (or (M2), with $\omega^2 > 0$ the extra term shown):*

$$\text{Var}(M) = \frac{v + \tau^2}{I} + \frac{\sigma^2}{IJ} \left[+ \frac{\omega^2}{J} \right].$$

Proof. $M = \frac{1}{IJ} \sum_{i,j} (\theta_i + u_i + w_j + e_{ij}) = \bar{\theta} + \bar{u} + \bar{w} + \bar{e}$, where $\bar{\theta} = \frac{1}{I} \sum_i \theta_i$, $\bar{u} = \frac{1}{I} \sum_i u_i$, $\bar{w} = \frac{1}{J} \sum_j w_j$ and $\bar{e} = \frac{1}{IJ} \sum_{i,j} e_{ij}$. The four components are independent and mean zero, so the variances add:

$$\text{Var}(M) = \frac{v}{I} + \frac{\tau^2}{I} + \frac{\omega^2}{J} + \frac{\sigma^2}{IJ}.$$

Note the asymmetry that does all the work: $\bar{u}$ averages I terms because u is indexed by i ; $\bar{w}$ averages J terms because w is indexed by j ; $\bar{e}$ averages IJ terms because e is indexed by both. The index set of a variance component dictates which count shrinks it. $\square$

Corollary 8 (J cannot reach the floor). *With $\omega^2 = 0$, $\inf_J \text{Var}(M) = (v + \tau^2)/I > 0$, and the total variance still purchasable by growing the panel from J to ∞ is exactly $\sigma^2/(IJ)$.*

Corollary 9 (I has no floor). *With $\omega^2 = 0$, $\text{Var}(M) \rightarrow 0$ as $I \rightarrow \infty$ for every fixed J , including $J = 1$.*

Corollaries 8 and 9 are the asymmetry, and together with Corollary 5 they are the argument: the item is the unit of replication. Note carefully that Corollary 9 is the half that *needs* $\omega^2 = 0$; see §6.

4.1 The exchange rate

Theorem 10 (Fixed-budget optimum). *Fix a budget of $B = IJ$ total judgments. Under (M1),*

$$\text{Var}(M) = \frac{J(v + \tau^2) + \sigma^2}{B},$$

*which is strictly increasing in J whenever $v + \tau^2 > 0$. Hence $J = 1$ — one judge on B items — minimises $\text{Var}(M)$, **unconditionally**. Under (M2), $\text{Var}(M) = \frac{J(v + \tau^2) + \sigma^2}{B} + \frac{\omega^2}{J}$ is strictly convex in J with interior minimiser*

$$J^* = \sqrt{\frac{\omega^2 B}{v + \tau^2}}.$$

Proof. Substitute $I = B/J$ into Theorem 7: $\text{Var}(M) = \frac{J}{B}(v + \tau^2 + \frac{\sigma^2}{J}) = \frac{J(v + \tau^2) + \sigma^2}{B}$, whose J -derivative is $(v + \tau^2)/B > 0$. With the ω^2/J term, the derivative is $(v + \tau^2)/B - \omega^2/J^2$, vanishing at J^* , and the second derivative $2\omega^2/J^3 > 0$ confirms a minimum. (For integer J , take the better of $\lfloor J^* \rfloor, \lceil J^* \rceil$.) $\square$

Theorem 10 is the statement I should have sent you in the first place. It is unconditional, it is a *design* rule rather than a limit statement, and it survives dropping v (if the I items are a fixed target set rather than a sample, replace $v + \tau^2$ by τ^2 throughout; the derivative is still positive).

Remark 11 (The version I sent, and which way your data falls). I previously framed this as: infinitely many judges on I items beats one judge on $2I$ items iff $v + \tau^2 < \sigma^2$. That is true — $\frac{v + \tau^2}{I} < \frac{v + \tau^2 + \sigma^2}{2I} \iff v + \tau^2 < \sigma^2$ — but it compares two *different* budgets (∞ against $2I$), which is why it is conditional. It is also *framing-dependent*, and I owe you both numbers. On the parameters implied by your 1.21/1.99 pair (§8, $J \approx 100$):

framing	quantity	vs σ^2	winner
items sampled (v counts)	$v + \tau^2 = 4.70$	> 1.00	one judge on $2I$ items, by $4.7\times$
items fixed (v drops)	$\tau^2 = 0.99$	< 1.00	infinitely many judges, by 1%

In the fixed-item framing the break-even is exactly $\bar{\varphi} = 1/2$, and your $\bar{\varphi} = 0.4975$ sits on the knife edge. So I will *not* sell you the unequal-budget comparison: in one of the two framings it is a coin flip on your own data. Use Theorem 10 instead, which is unconditional in both framings.

5 Which functional has the design effect (and two withdrawals)

I sent you a Proposition 4 claiming that the aggregate and per-item estimands have n_{eff} s differing by about a factor of four. That is wrong, and the error matters because you were going to lean on it. Here is the corrected picture.

Theorem 12 (One ceiling for estimation). *Under (M1), as J grows from 1 to ∞ :*

functional	value at J	limit	design effect
error variance of M	$(\tau^2 + \sigma^2/J)/I$	τ^2/I	$1/\bar{\varphi}$
per-item MSE, $\mathbb{E}(\bar{X}_i - \theta_i)^2$	$\tau^2 + \sigma^2/J$	τ^2	$1/\bar{\varphi}$
total $\text{Var}(M)$	$(v + \tau^2 + \sigma^2/J)/I$	$(v + \tau^2)/I$	$\leq 1/\bar{\varphi}$
$\text{Var}(\bar{X}_i F_i)$	σ^2/J	0	unbounded
detection, $\Pr(\text{all } J \text{ miss})$	$\mathbb{E}[p_i^J]$	0	unbounded

In particular the aggregate and the per-item estimation functionals have the same ceiling $1/\bar{\varphi}$.

Proof. The error of M is $\bar{u} + \bar{e}$ with variance $\tau^2/I + \sigma^2/(IJ)$; dividing the $J = 1$ value by the limit gives $(\tau^2 + \sigma^2)/\tau^2 = 1/\bar{\varphi}$ by Theorem 3. For a single item, $\bar{X}_i - \theta_i = u_i + \bar{e}_i$ has mean square $\tau^2 + \sigma^2/J$, the same ratio. For total $\text{Var}(M)$ the ratio is $(v + \tau^2 + \sigma^2)/(v + \tau^2)$, which is $\leq 1/\bar{\varphi}$ with equality iff $v = 0$. Conditioning on F_i removes u_i , leaving $\text{Var}(\bar{X}_i | F_i) = \sigma^2/J \rightarrow 0$. For detection in (M3), conditional independence gives $\Pr(\text{all miss} | p_i) = p_i^J$, so the marginal is $\mathbb{E}[p_i^J] \rightarrow 0$ by dominated convergence whenever $\Pr(p_i = 1) = 0$. $\square$

Withdrawal 1. My “per-item n_{eff} is nearly J ” rested on $\text{Var}(\bar{X}_i | F_i) = \sigma^2/J$. Conditioning on the item does make that go to zero — but conditioning removes u_i , and u_i is the error. What is left measures the panel’s precision about its own biased consensus, which is nobody’s estimand. For accuracy about the truth, per-item and aggregate share one ceiling. So “same panel, same data, two n_{eff} s differing by four” is withdrawn.

Withdrawal 2. I also wrote that the headline “implies 7/9” of savings is wrong by being too small. The correct contrast is sharper and in the opposite direction (§8): the headline invites a 7/9 cut that *costs* error variance.

What survives, and it is the part you need. Your observed divergence — aggregate ≈ 2 , per-item detection $\approx n$ — is real, and Theorem 12 locates it precisely: it is **estimation versus detection**, not aggregate versus per-item. And the mechanism is Corollary 6: τ^2 is a bias. Bias caps estimation, because averaging cannot move an offset. Bias does *not* cap detection, because a disjunction over judges only needs one judge on the right side of a threshold, and conditional independence — the thing you measured — is fully intact for that purpose. Your article already draws the variance/detection distinction; this is its mechanism, and it means your calibrating sentence was the right fix and should be promoted, not hedged.

Remark 13 (The decay rate). Detection has no floor but it does degrade. For $p_i \sim \text{Beta}(a, b)$, $\mathbb{E}[p_i^J] = B(a + J, b)/B(a, b) \sim (\Gamma(a + b)/\Gamma(a)) J^{-b}$: *polynomial* in J rather than the geometric $\bar{p}^J$ a naive independence calculation predicts. So item difficulty does cost you something in detection too — it converts a geometric rate into a polynomial one. That is worth a clause, because “no ceiling” is not the same as “no damage.”

6 The enemy: a judge main effect

This is the section Remark 2 promised, and the one place where your re-attribution is not yet safe.

Theorem 14 ($\bar{\varphi}$ is blind to ω^2 ; $\text{Var}(M)$ is not). *Work in* (M2): $X_{ij} = \theta_i + u_i + w_j + e_{ij}$, $\text{Var}(w) = \omega^2$.

(a) *The estimator in your recipe — the mean off-diagonal entry of `np.corrcoef` on the error columns — targets $\bar{\varphi}^{\text{cond}} = \tau^2/(\tau^2 + \sigma^2)$, free of ω^2 .*

(b) *Kish’s n_{eff} built from it is nonetheless the correct design effect for the error variance of the panel mean on an item, conditionally on the realised panel:*

$$\frac{\text{Var}(E_{ij} | w)}{\text{Var}(\bar{E}_i | w)} = \frac{\tau^2 + \sigma^2}{\tau^2 + \sigma^2/J} = \frac{J}{1 + (J - 1)\bar{\varphi}^{\text{cond}}}.$$

(c) But for a panel re-drawn from a judge population, $\text{Var}(M) = \frac{v+\tau^2}{I} + \frac{\sigma^2}{J} + \frac{\omega^2}{J}$, so $\inf_I \text{Var}(M) = \omega^2/J > 0$: **Corollary 9 is false when $\omega^2 > 0$.** For a fixed panel, $\bar{w}$ is instead a constant bias of M , equally unremovable by I .

Proof. (a) Pearson correlation between columns j and j' is computed across items and centres each column at its own mean. Conditionally on $(w_j, w_{j'})$ those are constants in i , so $\text{Cov}_i(E_{ij}, E_{ij'} | w) = \text{Var}(u_i) = \tau^2$ and $\text{Var}_i(E_{ij} | w) = \tau^2 + \sigma^2$: a per-judge additive shift is annihilated by the centring. (b) $\bar{E}_i | w = u_i + \bar{w} + \bar{e}_i$, so $\text{Var}(\bar{E}_i | w) = \tau^2 + \sigma^2/J$; divide and simplify by $(\tau^2 + \sigma^2)$. (c) Immediate from Theorem 7, noting that $\bar{w}$ averages J terms and is constant across i , so no amount of I touches it. $\square$

The structure of the danger is worth stating plainly, because it is a shape I keep meeting: **the instrument you ran cannot see the quantity that would refute the conclusion you drew from it.** $\hat{\varphi} = 0.49$ is compatible with $\omega^2 = 0$ and with $\omega^2 = 25$ — I checked, and $\hat{\varphi}$ reads 0.5001 in both cases. So “we measured $\hat{\varphi}$ and the co-failure is all item-level” does not license “therefore nothing is judge-attributable.” ω^2 is exactly the judge-attributable part, and it is in your data’s blind spot.

What to measure, concretely. ω^2 is the between-judge variance of per-judge mean error — the thing your $\hat{\varphi}$ throws away when it centres the columns. Take your error matrix $\mathbf{E}$ (items $\times$ judges) and compute the one-way ANOVA across columns:

$$\hat{\omega}^2 = \text{Var}_j(\text{column mean of } \mathbf{E}) - \frac{\widehat{\tau^2 + \sigma^2}}{I},$$

the subtraction removing the sampling noise in each column mean. Then compare $\hat{\omega}^2$ to τ^2 . Three outcomes, three different headlines:

- $\hat{\omega}^2 \approx 0$: the re-attribution is clean and unconditional. “The item is the unit of replication.”
- $\hat{\omega}^2 \ll \tau^2$ but non-zero: re-attribution holds with a stated remainder, and $J^* = \sqrt{\omega^2 B / (v + \tau^2)}$ from Theorem 10 gives the right panel size rather than $J = 1$.
- $\hat{\omega}^2 \gtrsim \tau^2$: the ceiling is *shared*, not relocated. Say so.

I cannot run this: your repository at 3fa2760 contains the draft, the L^AT_EX and the build script, but no alphaNLI data. That is the one gap in this document that is yours to close.

7 ESDOF: errors, and why

You asked whether to keep ESDOF on raw scores. **Errors.** But your stated reason — consistency with the Kish function — is the weaker one; the real reason is identifiability, and there is a second finding here that I think you will want before publishing.

Theorem 15 (ESDOF: raw scores measure signal, errors measure $\bar{\varphi}$). *Let $\text{PR}(C) = (\sum_k \lambda_k)^2 / \sum_k \lambda_k^2$ for the eigenvalues of C . Under (M1):*

(a) *The judge $\times$ judge covariance of raw scores is $(v + \tau^2)\mathbf{11}^\top + \sigma^2\text{Id}$, so*

$$\text{ESDOF}_{\text{raw}} = \frac{J^2(a + \sigma^2)^2}{(Ja + \sigma^2)^2 + (J - 1)\sigma^4}, \quad a := v + \tau^2,$$

a function of a/σ^2 only. It equals J when $a = 0$ and equals 1 **when** $\sigma^2 = 0$ — that is, it is minimised exactly when the judges are perfect.

(b) On errors, $\text{ESDOF}_{\text{err}}$ is the same formula with $a = \tau^2$, hence a function of $(J, \bar{\varphi})$ alone, strictly decreasing in $\bar{\varphi}$ from J to 1.

(c) The two statistics you print as n_{eff} have different ceilings:

$$\lim_{J \rightarrow \infty} \text{ESDOF}_{\text{err}} = \frac{1}{\bar{\varphi}^2}, \quad \lim_{J \rightarrow \infty} n_{\text{eff}}^{\text{Kish}} = \frac{1}{\bar{\varphi}}.$$

Proof. (a),(b) A compound-symmetric $a\mathbf{1}\mathbf{1}^\top + s\text{Id}$ has eigenvalue $Ja + s$ on $\mathbf{1}$ and s with multiplicity $J - 1$; substitute into PR. For raw scores $\text{Cov}(X_{ij}, X_{ij'}) = v + \tau^2$ and $\text{Var}(X_{ij}) = v + \tau^2 + \sigma^2$. Setting $\sigma^2 = 0$ gives $J^2 a^2 / (J^2 a^2) = 1$; setting $a = 0$ gives $J^2 s^2 / (Js^2) = J$. (c) Normalise $\tau^2 + \sigma^2 = 1$, so $\tau^2 = \bar{\varphi}$ and $\sigma^2 = 1 - \bar{\varphi}$. Then $\sum_k \lambda_k = J\bar{\varphi} + 1 - \bar{\varphi} + (J - 1)(1 - \bar{\varphi}) = J$ exactly, while $\sum_k \lambda_k^2 = (J\bar{\varphi} + 1 - \bar{\varphi})^2 + (J - 1)(1 - \bar{\varphi})^2 = J^2 \bar{\varphi}^2 + O(J)$. Hence $\text{PR} = J^2 / (J^2 \bar{\varphi}^2 + O(J)) \rightarrow 1/\bar{\varphi}^2$. $\square$

Why errors (the identifiability argument). By (a), raw-score ESDOF is a reparametrisation of the signal-to-noise ratio $(v + \tau^2)/\sigma^2$. The θ_i are shared across judges *by construction* — agreeing about θ is the entire job. So a low raw-score ESDOF is a report of *signal*, not of unreliability, and part (a)’s degenerate case makes this maximally sharp: a panel of *perfect* judges has $\text{ESDOF}_{\text{raw}} = 1$ exactly. A reliability statistic minimised by perfect judges is measuring the wrong thing. Your switch was right; argue it this way instead of “for consistency,” because consistency would be equally served by moving Kish onto raw scores, which would be wrong.

The finding you should act on. By (c), ESDOF and Kish are *not* two routes to one number. On your panel at $\bar{\varphi} = 0.49$ their ceilings are 4.16 and 2.04, and at $J = 9$ they read 3.08 and 1.83 — a 68% disagreement. Your text says “ESDOF screens a design; Kish audits a deployment,” which is a good division of labour, but both functions are named `n_eff` and both are glossed as “the effective number of independent judges,” and a reader who runs the snippet will get two numbers and no way to reconcile them. Theorem 12 settles which is which: **Kish is the variance effective- n** — it is literally the design-effect ratio — **and ESDOF is not a variance effective- n at all**. The participation ratio counts effective *dimensions* of the correlation spectrum; it does not appear in any variance formula for M or $\bar{X}_i$. I would rename it (`spectral_dimension`, or `esdof`, but not `n_eff`) and say in one clause that it is not on the same scale as Kish.

A small bug. Both of your functions return `nan`, silently, on a panel with no error variance: `keep = E.std(axis=0) > 0` drops every column, `np.corrcoef` gets a 0-column array and the ratio becomes 0/0. A perfect panel is exactly the input a reader will try first on toy data, and `nan` propagates into a dashboard without complaint. Guard it and return J (or raise).

8 Your numbers

You report three things: $\approx 49\%$ of error variance is between-item; raw-score $\bar{\varphi}$ gives $n_{\text{eff}} = 1.21$; error-based gives $n_{\text{eff}} = 1.99$. Theorem 3 says the first *is* $\bar{\varphi}$, so these are over-identified and can be checked against each other.

Consistency, and the panel size. Inverting Kish ($J = t(1 - p)/(1 - tp)$ for $n_{\text{eff}} = t$) at $p = 0.49$ gives

$$n_{\text{eff}} = 1.99 \iff J \approx 41.$$

At $J = 9$, $\bar{\varphi} = 0.49$ yields $n_{\text{eff}} = 1.829$; conversely $n_{\text{eff}} = 1.99$ at $J = 9$ requires $\bar{\varphi} = 0.4403$, which contradicts your 49%. So the 1.99 and the 49% are mutually consistent only if the alphaNLI panel has $J \gtrsim 40$ members. If it is the full ChaosNLI annotator set ($J = 100$), then $\bar{\varphi} = 0.4975$ — which rounds to your 49%, and is the reading I have used throughout. Either way, with σ^2 normalised to 1:

$$J = 41 : (v, \tau^2, \sigma^2) = (3.66, 0.96, 1.00); \quad J = 100 : (v, \tau^2, \sigma^2) = (3.71, 0.99, 1.00).$$

These reproduce 1.21 and 1.99 simultaneously, and the 49% comes out as a prediction rather than a fit. **Please confirm J** — it is the one input I inferred rather than read.

The 1.99 is the ceiling. At $\bar{\varphi} = 0.49$ the ceiling is 2.041 and 1.99 is 97.5% of it. This is not “nine judges gave us two.” It is saturation. Hence verdict item 2: state the ceiling.

What a panel cut would cost. At $(v, \tau^2, \sigma^2) = (3.71, 0.99, 1.00)$:

error variance of M , $J = 9 \rightarrow \infty$	1.101 $\rightarrow$ 0.990, i.e. 10.1% purchasable
total $\text{Var}(M)$, $J = 9 \rightarrow \infty$	4.801 $\rightarrow$ 4.690, i.e. 2.3% purchasable
error variance of M , $J = 9 \rightarrow 2$	1.101 $\rightarrow$ 1.490, i.e. + 35.3%

A reader who takes “paying for nine, getting two” at face value cuts to two, banks 7/9 of the spend, and buys a third more error. The number is right; the *action* it suggests is backwards. That is the strongest practical reason to re-scope rather than retract.

An object substitution in the text. At 3fa2760 the sentence reads “correlating raw label indices *on this panel* gives $n_{\text{eff}} \approx 1.2$... the error-based version gives ≈ 2.0 .” Per UID 742 these are measured on ChaosNLI alphaNLI. But `grep -ci 'alphanli|chaosnli'` returns 0 in both `_article.tex` and `DRAFT-2026-09-11.md`: the dataset is never named in the article. The nearest antecedent for “this panel” is Kohli’s nine frontier LLM judges from the opening. So a crowd-annotator measurement on an NLI benchmark reads as a measurement on the nine-LLM panel — and because 2.0 and Kohli’s 2.18 agree in magnitude, nothing in the text makes the substitution visible. Name the dataset and the J in that sentence. (By the arithmetic above the two panels cannot be the same object: $\bar{\varphi} = 0.49$ at $J = 9$ gives 1.83, not 1.99.)

9 What I would change in the article

1. **Keep the number. Promote it to a ceiling.** “No panel size buys more than about two” replaces “nine buys two.” Corollary 4.
2. **Add one sentence of attribution.** “And the thing that caps it is the item, not the judges” — with Corollary 5 as the reason: the numerator of $\bar{\varphi}$ is the variance of an item functional.
3. **Add the design rule.** Theorem 10: at a fixed judgment budget, fewer judges on more items strictly wins. This is the actionable replacement for the panel cut a reader would otherwise make.

4. **Name the dataset and J** in the 1.2/2.0 sentence (§8).
5. **Say “bias,” not “redundancy.”** Corollary 6 is the one-line explanation of “everyone finds the hard items hard,” and it is what makes the ceiling feel inevitable rather than surprising.
6. **Keep ESDOF on errors, re-argue it from identifiability,** stop calling it n_{eff} , and guard the nan (§7).
7. **Hold the re-attribution until $\hat{\omega}^2$ is measured** (§6). This is the only item on this list that is a blocker, and it is a twenty-minute computation.

10 Verification

All of the above is checked in `proofs/code-2026-10-01/verify_unit_of_replication.py` (14 tests). The estimators are copied verbatim from `DRAFT-2026-09-11.md` — the propositions are tested against the code you ship, not against my own idealisation of it.

test	object	result
1	Thm 3, (M1)	8 parameter sets; worst $ \hat{\varphi} - \tau^2/(\tau^2 + \sigma^2) = 1.8 \times 10^{-3}$ at $I = 4 \times 10^5$ (Monte-Carlo scale $I^{-1/2} = 1.6 \times 10^{-3}$)
7	Thm 3, (M3)	6 laws for p_i (uniform, skew, bimodal, tight, two-point, degenerate) $\times J \in \{9, 100\}$; worst error 6.9×10^{-4} . No normality used
2	negative control	$\tau^2 = 0 \Rightarrow \hat{\varphi} \in [-1.2, 1.0] \times 10^{-3}$ and $n_{\text{eff}} = 2.002, 8.993, 25.09, 99.73$ at $J = 2, 9, 25, 100$: the ceiling evaporates. Positive control at $\tau^2 = 1$, same (v, σ^2) , reports 2.000
11	Thm 7	symbolic expansion of $\mathbb{E}[M^2]$ minus the closed form = 0 exactly at $(I, J) = (3, 4), (1, 1), (2, 7), (5, 3), (9, 9)$, including the ω^2 term
12	birth-range break	4000 random draws, 4 decades per parameter: Prop 2 identity 4000/4000 (worst rel. err. 3.7×10^{-16}); Kish ceiling strict 4000/4000; budget monotonicity 4000/4000
6a	Thm 14(a)	$\hat{\varphi} = 0.5001$ for $\omega^2 \in \{0, 0.25, 1, 4, 25\}$, against $\tau^2/(\tau^2 + \omega^2 + \sigma^2) \in \{0.50, 0.44, 0.33, 0.17, 0.04\}$
6b,c	Thm 14(c)	$\widehat{\text{Var}}(M)$ tracks base + ω^2/J and does not shrink in I ; integer $\arg \min_J$ matches J^* (7 vs 6.71; 21 vs 21.21)
8,9	Thm 15	closed forms to 4 d.p.; $\text{ESDOF}_{\text{raw}}$ moves $3.00 \rightarrow 1.018$ as $v : 0 \rightarrow 100$ at fixed (τ^2, σ^2) while $\text{ESDOF}_{\text{err}}$ stays 3.000; $\text{ESDOF}(10^6)$ vs $1/\bar{\varphi}^2$ agree at $\bar{\varphi} \in \{0.1, 0.25, 0.49, 0.4975, 0.64, 0.9\}$; both of your functions return nan on a perfect panel
13,14	Thm 12	$\text{DE}[\text{err of } M] = \text{DE}[\text{per-item MSE}] = 1/\bar{\varphi}$ exactly on 4 parameter sets while $\text{DE}[\text{Var}(M)]$ ranges 1.01–2.00; $\mathbb{E}[p_i^J] \sim J^{-b}$ for $p_i \sim \text{Beta}(a, b)$
4b	instrument check	the uniform 2% shortfall in an earlier Monte-Carlo pass was one shared seed block reused across rows; over 4 independent blocks the ratio straddles 1 (0.980, 0.997, 1.015, 1.018), and at 6×10^4 replicates it is 1.005 against a relative standard error of 0.006

11 Gaps

Stated precisely, because an incomplete proof with a located gap is worth more than a complete one with a hidden assumption.

1. **ω^2 is unmeasured, and it is load-bearing for exactly one claim.** Corollary 9 (“ I has no floor”) is *false* if $\omega^2 > 0$, and Theorem 14(a) shows your $\hat{\hat{\varphi}}$ cannot detect ω^2 . Theorems 3, 7, 10, 12 and 15 are unaffected — in particular *the number survives ω^2 unconditionally*. This is the only blocker.
2. **J for the alphaNLI panel is inferred, not read.** §8 shows the 49%/1.21/1.99 triple is consistent only for $J \gtrsim 40$ and I have assumed $J = 100$. Not in the repository at 3fa2760. The qualitative conclusions do not depend on it; the parameter table does.
3. **(ECI)(i) is measured on one dataset.** You report residual correlation ≈ 0 after per-item demeaning on alphaNLI. I take that as given. On a panel where it fails, $\bar{\varphi}$ splits into an item part and a genuine shared-blind-spot part, and the attribution is correspondingly split. Your article’s own “upper bound on genuine co-failure” caveat is the right hedge and should stay.
4. **Everything here is second-moment.** Theorems 7–12 optimise variance and MSE. They say nothing about adversarial robustness, prompt-injection resistance, or the value of a panel as a tripwire — all of which are real reasons to keep more judges than J^* , and none of which this document models. Theorem 12’s detection row is the nearest thing, and it points the *other* way.
5. **I am not a measurement theorist.** $\bar{\varphi}$, n_{eff} , ESDOF and the e-process are yours. What I can defend is that (ECI) implies everything above and that (ECI)(i) is what you measured. Whether (ECI)(ii) is the right idealisation of a judge panel is your call, and §6 is the test I would run before making it.